

Project Initialization and Planning Phase

Date	2 JULY 2026
Team ID	
Project Title	Flood Prediction Using Machine Learning
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction by using Machine Learning techniques to analyze historical weather data and predict the likelihood of flood events. The system enhances the accuracy and speed of flood forecasting, enabling authorities to issue timely warnings and reduce the impact of natural disasters. It integrates the trained model into a Flask-based web application, providing an easy-to-use interface for disaster management officials. The application is designed for deployment on IBM Cloud, ensuring scalability and accessibility.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based Flood Prediction System that accurately predicts flood occurrence using historical weather data, enabling early warnings and supporting disaster management authorities in making timely decisions.
Scope	The project focuses on collecting and analyzing historical weather data such as annual rainfall, cloud visibility, and seasonal rainfall patterns. Multiple Machine Learning algorithms—Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and XGBoost—are trained and evaluated to identify the best-performing model. The selected model is deployed as a Flask web application for real-time flood prediction.
Problem Statement	
Description	Existing flood forecasting methods often fail to provide timely and accurate predictions due to changing weather patterns and reliance on traditional techniques. This results in delayed warnings, insufficient disaster preparedness, and increased damage to life and property.
Impact	An accurate flood prediction system enables early evacuation planning, efficient resource allocation, and faster emergency response. It helps reduce the loss of lives, minimizes property damage, and improves disaster management through reliable and timely flood risk assessment.
Proposed Solution	

Approach	Develop a Machine Learning-powered flood prediction model using historical weather data. The system preprocesses the data, trains multiple classification algorithms (Decision Tree, Random Forest, KNN, and XGBoost), selects the best-performing model, and deploys it through a Flask web application for real-time flood risk prediction.
Key Features	☐ Predicts flood occurrence using historical weather data. ☐ Uses Decision Tree, Random Forest, KNN, and XGBoost algorithms. ☐ Achieves 96.55% accuracy using the XGBoost model. ☐ Provides a simple Flask web interface for entering weather data. ☐ Supports early warning and evacuation planning. ☐ Can be deployed on IBM Cloud for remote access and scalability.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU for model training	Intel Core i5/i7 or equivalent, optional GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models.	500 GB SSD or above
Software		
Frameworks	Backend Framework	Flask
Libraries	Machine Learning & Data Processing	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Historical Weather Dataset	CSV format containing rainfall, cloud visibility, seasonal rainfall, and flood occurrence records